



School:Campus:

Academic Year: Subject Name: Subject Code:

Semester: Program: Branch: Specialization:

Date:

Applied and Action Learning

(Learning by Doing and Discovery)

Name of the Experiment: Mint it Yourself – NFT Creation and Deployment

*Objectives/Aim:

The objective of this experiment is to understand and implement the complete lifecycle of a non-fungible token (NFT) by creating its metadata, hosting it on IPFS through Pinata, deploying an ERC-721 smart contract using Remix IDE, and minting the NFT to a wallet address with the linked metadata.

*Coding Phase: Pseudo Code / Flow Chart / Algorithm

1. Create a metadata.json file on your system with the keys name, description, image, and attributes.
2. Log in to your Pinata Cloud account, upload a .jpg or .png file, and copy its URL.
3. In the metadata.json file, paste the copied image URL in the value of the image key and save the file.
4. Upload the metadata.json file to Pinata, open it, and copy the generated metadata URI.
5. Open Remix IDE, create a new Solidity smart contract, and compile it.
6. After successful compilation, deploy the contract by selecting Injected Provider - MetaMask as the environment, and initialize it with the name ("Apecollection"), symbol ("APEC"), and the initial owner (your MetaMask wallet address).
7. Deploy the contract by confirming the transaction through the MetaMask popup.
8. After successful deployment, go to the Deployed Contracts section and find the MINTTO function.
9. In the MINTTO section, enter your wallet address in the "to" field and paste the Pinata metadata.json URI in the "metadataURI" field.
10. Click on Transact and confirm the transaction in the MetaMask popup.
11. Once the transaction is confirmed, your NFT will be minted successfully.
12. Open your MetaMask wallet and check the NFTs section to view your newly minted token.

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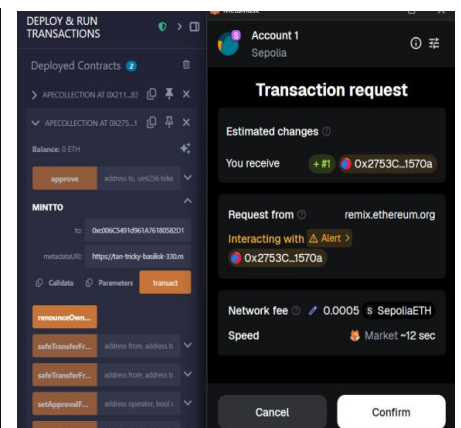
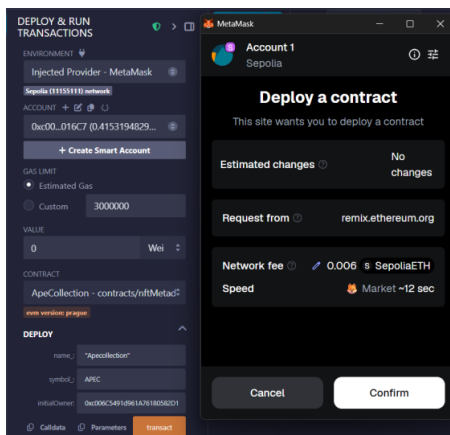
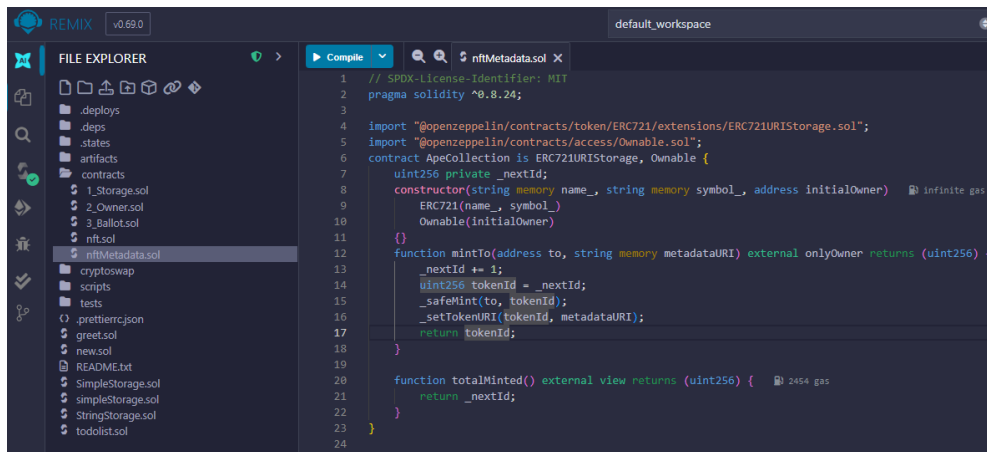
*** As applicable according to the experiment.
Two sheets per experiment (10-20) to be used.**

* Testing Phase: Compilation of Code (error detection)

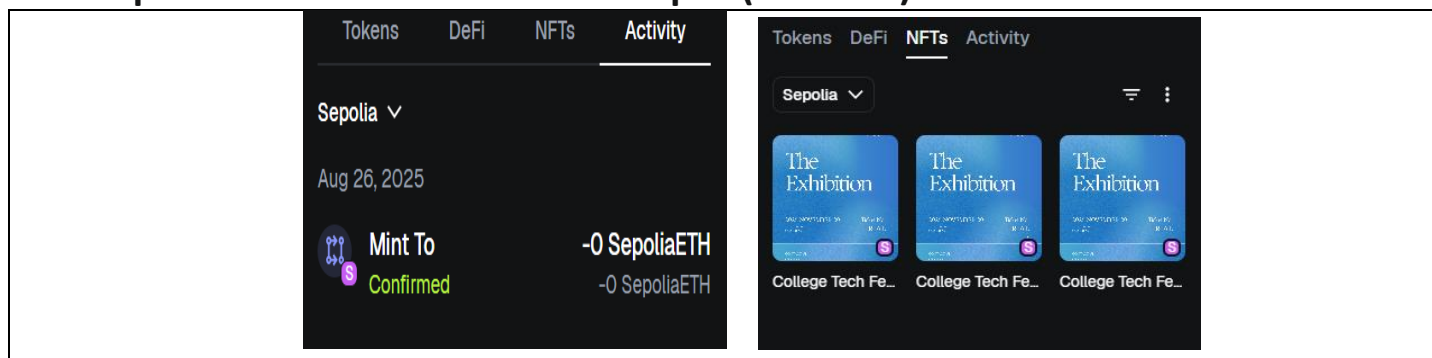
NO ERROR

* Implementation Phase: Final Output (no error)

```
{
  "name": "CUTM Badge #2",
  "description": "NFT demo for Blockchain Studnets on Sepolia.",
  "image": "https://tan-tricky-basilisk-330.mypinata.cloud/ipfs/bafybeibug46wz2inmk6hqprfyvq5e3n15kuzr5y5jdiwxzh3psy3e5pvnu",
  "attributes": [
    {
      "trait_type": "Department",
      "value": "CSE"
    },
    {
      "trait_type": "Campus",
      "value": "BBSR"
    }
  ]
}
```



* Implementation Phase: Final Output (no error)



* Observation :

From this experiment we observed that :

- The NFT creation process integrates both off-chain (Pinata) and on-chain (smart contract) components.
- Manual metadata and URI management are essential steps before minting.
- Remix IDE and MetaMask provide a beginner-friendly environment for smart contract deployment.
- Pinata ensures reliable and decentralized storage of NFT assets.
- Successful NFT minting requires accurate input of wallet address and metadata URI.
- MetaMask acts as both a deployment tool and wallet for holding the minted NFTs.
- The entire process demonstrates a clear workflow for self-minting NFTs without third-party platforms.

ASSESSMENT

Rubrics	Full Mark	Marks Obtained	Remarks
Concept	10		
Planning and Execution/Practical Simulation/ Programming	10		
Result and Interpretation	10		
Record of Applied and Action Learning	10		
Viva	10		
Total	50		

Signature of the Student :

Name :

Signature of the Faculty :

Regn. No. :

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